

Example 1: Study Habit vs Grade

2×3 table (n = 180). Compute expected counts, $\chi^2 \approx 9.6$, df =

2 $\Rightarrow$ reject H_0 at $\alpha = 0.05$; active study associated with higher grades.

Example 2: Device Type vs Purchase

3×2 table (n = 150). $\chi^2 \approx 4.1$, df = 2 $\Rightarrow$ fail to reject at α

= 0.05; insufficient evidence of association.